

Lennox Magak

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Building Toward DeFi and Blockchain Engineering | Open to SWE, Full-Stack, and Data Engineering Roles

EDUCATION

Concordia College, Moorhead, MN

Graduated: May 3rd 2026

Bachelor of Arts in Computer Science, Minor in Data Analytics

PROJECTS & EXPERIENCES

Predicting County-Level Small Business Formation in the United States / *Sole Researcher* / Jan, Feb, Mar, Apr 2026

- A predictive research project asking which county-level demographic and economic characteristics best predict the rate of new business formation across all 3,069 US counties, and how accurately a county's startup activity can be predicted from its demographic profile alone.
- Constructed the modelling table directly from federal sources: the US Census Business Dynamics Statistics 2023 release (target variable, establs_entry per 1,000 residents) joined on 5-digit FIPS to American Community Survey 5-Year Estimates 2019 to 2023 (six predictors covering income, education, foreign-born share, labour-force participation, mean commute, and prime-age population share).
- Pulled the ACS slice through the Census Data API at api.census.gov/data/2023/acs/acs5; downloaded the BDS county time-series CSV directly from www2.census.gov; documented the full acquisition pipeline so the project is reproducible.
- Fitted competing model families against the target and discussed which county-level levers actually move the predicted startup rate; framed the use case so that the Small Business Administration, a state economic-development office, or an academic researcher could each use the model for a distinct policy question.
- Wrote up the project in an Rmarkdown report with paired DataPull and Modelling notebooks; rendered HTML, Word, and slide deliverables.

Technologies Used: R, Rmarkdown, Census BDS, American Community Survey API, regression and tree-based models, reproducible analysis pipelines

GitHub: <https://github.com/Lennox7384/data318-modeling>

Project RailGuard: Safety-Critical Railway Crossing Simulation / *Software Engineer & Hazard Analyst* / Nov, Dec 2025

- Co-built RailGuard, a single-track automated railway crossing simulation modeling a high-stakes environment where a missed gate closure or a malicious sensor command could mean catastrophic loss; the project shifted engineering focus from algorithmic complexity to process integrity.
- Owned the hazard analysis track: produced a Fault Tree Analysis diagram for the "Gate Open" failure mode, defined mitigation strategies for identified hazards, and helped drive the team to a formal Waterfall lifecycle (planning, requirements, design, implementation, verification, postmortem) rather than ad-hoc coding.
- Connected the React UI to the Boolean state-machine controller (RailGuardController.js, Sensor.js) using useEffect hooks, then wrote integration touch-points so unit tests in vitest could exercise sensor inputs and gate transitions in isolation.
- Demonstrated end-to-end ownership of a system where requirements traceability, failure-mode reasoning, and verification rigor were graded as heavily as code; a profile aligned with safety-critical, fintech, and regulated software roles.

Technologies Used: React, Vite, JavaScript, vitest, ESLint, Fault Tree Analysis, State Machine pattern, Waterfall SDLC

GitHub: <https://github.com/Lennox7384/project-railguard>

NLP Module for Multi-Component AI Pipeline / *Sole Developer* / *Sep, Oct 2025*

- Designed the NLP module of a multi-component AI system; the module exposes three core functions (`preprocess_text`, `analyze_sentiment`, `extract_entities`) and a `process_text` wrapper that returns a JSON envelope downstream modules can consume.
- Implemented the full processing pipeline: input validation, tokenisation, NLTK VADER sentiment analysis with the standard ± 0.05 compound-score bands, and POS-tag-based noun entity extraction. The JSON envelope is designed as the stable contract between this module and three downstream consumers.
- Structured the JSON output so the same module feeds an Artificial Neural Network (sentiment classification training, feature extraction), a Text-to-Speech module (sentiment-conditioned voice tone, emphasis on named entities), and a Computer Vision pipeline (token-driven scene focus).
- Added defensive input validation, a standalone smoke-test block runnable via `python nlp_module.py`, and a pytest suite of 8 cases covering empty input, the three sentiment branches, entity extraction, and the JSON envelope contract.

Technologies Used: Python, NLTK (VADER, punkt, averaged_perceptron_tagger), JSON, regular expressions, modular integration design

GitHub: <https://github.com/Lennox7384/nlp-module>

Concordia College Entrepreneurship Club Website / *Full-Stack Web Developer* / *Sep, Oct, Nov, Dec 2024*

- Partnered with a four-person team to build a public-facing website for the Concordia College Entrepreneurship Club, surfacing membership, mission, activities, and an events history that the club previously tracked off-platform.
- Owned the Schedule (events) feature end-to-end: structured the markup in HTML, styled a responsive calendar and event list with CSS Flexbox, and wired a PHP backend against a MySQL database so past, present, and future events render dynamically through JavaScript fetches.
- Built data-driven UI components that update without a page reload; a pattern directly applicable to modern dashboards, content management surfaces, and any product where event data drives the view.
- Reinforced full-stack fluency across the LAMP-style stack and proved that I can take a real, ongoing organisation as a client and ship a working product into their hands.

Technologies Used: HTML, CSS (Flexbox), JavaScript, PHP, MySQL

GitHub: <https://github.com/Lennox7384/entrepreneurship-club-website>

Revolve Bank Application Software / *Project Manager & Lead Developer* / *Jan, Feb, Mar, Apr, May 2024*

- Architected a Java-based retail-banking simulation as project manager for a four-person team, designing the institutional hierarchy (Branch, Manager, Tellers, Customers) so domain responsibilities map cleanly to classes.
- Authored the `Branch`, `Manager`, and `Teller` classes; embedded financial calculations such as branch revenue and liquidity checks, account oversight operations like `freezeAll`, and the core transaction surface (`transferFunds`, `createNewCustomer`, `withdrawCustomerFunds`).
- Coordinated sprint-style milestones, code-review pairings, and integration testing across the team; the system shipped on schedule with the full banking surface running end-to-end.
- Advanced personal skills in object-oriented design, role-based aggregation, and financial system modelling; transferable foundations for fintech back-end engineering and the kinds of policy-rich systems that dominate enterprise SWE roles.

Technologies Used: Java, JUnit, object-oriented design

GitHub: <https://github.com/Lennox7384/revolve-bank-application>

Star Wars Characters Simulation / *Sole Developer* / *Jan, Feb 2024*

- Designed and implemented a Java-based simulation of Star Wars characters to demonstrate advanced object-oriented programming, focusing on the `Cloneable` interface across two key classes; first project where I shipped working deep-copy semantics in Java.

- Achieved full method coverage by leveraging the JUnit testing framework, validating 16 major methods with positive, negative, and edge cases to ensure robust performance; boundary-validation tests pin the contract on `Force.setStrength` (1 to 100) using `@Test(expected)` annotations.
- Strengthened skills in coding hygiene, structured problem decomposition, and software testing; an early proof point of test-driven discipline that I now carry into every project.

Technologies Used: Java, JUnit 4, Maven, object-oriented design

GitHub: <https://github.com/Lennox7384/star-wars-characters-simulation>

Covid-19 Spread Simulation System / *Sole Developer* / *Oct, Nov, Dec 2023*

- Developed a Java simulation of COVID-19 transmission across a 40x40 grid (1,600 cells) using a Susceptible, Infected, Recovered (SIR) model layered with probabilistic mechanics; ran simulations across 250 time steps with real-time statistical tracking.
- Engineered a heterogeneous population: base citizens spread infection (80% unmasked, 10% masked transmission rates), `CitizenType_A` used viral-load diagnostics to infect or eliminate neighbours, `CitizenType_B` introduced non-COVID population losses, and Doctor citizens treated infections with recovery probabilities (e.g., 80% on masked patients).
- Built a Swing GUI to visualise population dynamics in real time with colour-coded cell states (red for infected, blue for doctors, pink and magenta for events) and temporary flashes highlighting key interactions such as treatment or sudden disruptions.
- Modelled socio-economic and healthcare dynamics; privileged diagnostics, societal disruptions, and medical intervention; tracking infections, recoveries, COVID-related losses, and non-COVID reductions to analyse population trends, with summary statistics printed to the console at each step.

Technologies Used: Java, Java Swing, `java.util.Random` (probabilistic modelling), object-oriented design

GitHub: <https://github.com/Lennox7384/covid-19-spread-simulation>

Motor Vehicle Risk Factors Across US States / *Sole Analyst* / *Fall 2023, Python port May 2026*

- Produced a state-level analysis of the factors associated with motor-vehicle fatality risk across the United States; assembled a custom 50-state dataset from NHTSA fatality records, state-level alcohol consumption, average miles per driver, pickup-truck share, and per-capita police counts.
- Constructed appropriate per-capita rates so cross-state comparisons would not be confounded by population size; identified miles-per-driver as the strongest single predictor ($r = 0.6640$, $R^2 = 0.4409$) and a joint OLS regression on miles-per-driver and pickup-truck share achieving $R^2 = 0.4959$.
- Re-implemented the full analysis in Python (`pandas`, `statsmodels`, `matplotlib`, `scipy`) in May 2026 to make the methodology verifiable in code and reproducible from raw data files; published the rendered notebook to GitHub Pages so reviewers can read it without installing Python.
- Delivered both the original 2023 Excel report and the 2026 Python port in one repository; an end-to-end demonstration that I can ship a research-style data story and then upgrade it under modern tooling without losing the original narrative.

Technologies Used: Python (`pandas`, `statsmodels`, `scipy`, `matplotlib`), Jupyter, Excel (original analysis), state-level survey data integration

GitHub: <https://github.com/Lennox7384/motor-vehicle-fatality-analysis>